

H: Holy Network

Source file name: `holy.c`, `holy.cpp`, `holy.java`, or `holy.py`

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Once a century, when the stars align in a sacred conjunction, the world's greatest alchemists leave their secret laboratories to gather at the Arcane Conclave. Their purpose is to perform the Great Concordance Ritual, an act of magic so immense it can rewrite the laws of nature.

Each master brings a unique ingredient, distilled over a century of work. Through arcane numerology and the study of the vibrational patterns of matter, each alchemist has managed to assign their ingredient an integer non-negative number that defines its essence: the *Harmonic Resonance*.

The ritual, however, is dangerously unstable. For the power to manifest safely, the group of alchemists channeling it must be in perfect harmony. The rule is strict and ancient: a group is harmonic only if the Resonance of every pair of ingredients within the group is compatible. Two ingredients are *compatible* if their numbers descend from the same numerical root (i.e., they share a factor greater than 1).

Forming the largest possible group is crucial, as the ritual's power increases exponentially with each participant. As the new alchemist and expert in numerology for the Conclave, you have been entrusted with the crucial task of analyzing all the ingredients and determining the maximum size of the harmonic group that can be formed, thus ensuring the ritual's success.

The chronicles of the Conclave of the Faded Star, the last to be held, detail a particularly complex challenge. Eight masters presented ingredients with Harmonic Resonances of **30, 42, 70, 105, 11, 13, 17 and 19**. The chronicles state that after a long deliberation, it was discovered that the largest possible harmonic group was of size 4, formed by the resonances **30, 42, 70 and 105**. An incomplete verification of that, as it was recorded, was as follows:

- The pair (30, 42) is compatible, as they descend from the numerical root 6.
- The pair (42, 70) is compatible, as they descend from the root 14.
- The pair (105, 42) is compatible, as they descend from the root 21.
- ...

This discovery was a milestone in the Conclave's history, as it proved that the addition of the other prime ingredients could not expand the core harmonic group, and that true harmony resided in a complex web of shared roots.

Input

The input consists of multiple test cases. Each test case begins with a line containing a single integer N ($1 \leq N \leq 50$), the number of ingredients. The next line contains N space-separated integers $p_1, p_2, \dots, p_n$ ($1 \leq p_i \leq 10^{12}$), which are the Harmonic Resonances. The input terminates when a line with $N = 0$ is encountered, which should not be processed as a case.

The input must be read from standard input.

Output

For each test case, print a single line with a single integer: the size of the largest harmonic group that can be formed with the given Harmonic Resonances.

The output must be written to standard output.

Sample Input	Sample Output
8 30 42 70 105 11 13 17 19 5 30 42 70 105 154 5 2 3 5 7 11 0	4 5 1